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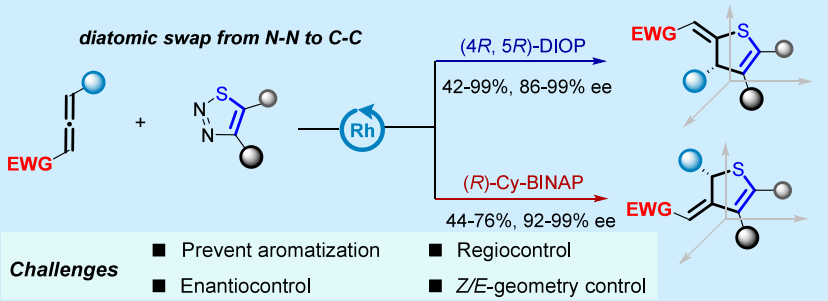
Enantioselective and Regiodivergent Skeletal Transformations of 1,2,3-Thiadiazoles to Alkylidene Dihydrothiophenes

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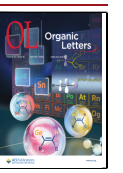
ABSTRACT: By leveraging the Rh-catalyzed asymmetric skeletal editing of 1,2,3-thiadiazoles, we successfully achieve the ligand-controlled enantioselective and regiodivergent synthesis of structurally diverse 2- and 3-alkylidene 2,3-dihydrothiophenes. Late-stage structural editing of complex natural and drug molecules is also enabled.

Thiacycles are prevalent and privileged pharmacores in many medicinally relevant molecules.[1] They are found in 128 US FDA-approved drugs, 18 best-sold drugs by retail sales,[2] and many other bioactive molecules. However, very little attention has been paid to alkylidene dihydrothiophenes (type I, type II, and type III), which possess 3-dimensional-structures and one stereocenter (Scheme 1A). These structural properties cater to the well-recognized concept—escape from flatland,[3] making them promising scaffolds in drug development campaigns. The enantioselective synthesis of these molecules nonetheless is highly challenging. The limited synthetic routes mainly relied on the dearomative functionaliztion of parent thiophene rings (Scheme 1B).[4−6] In 2021, Sun and Houk reported the chiral phosphoric acid (CPA)-catalyzed enantioselective dearomative functionalization of thiophenes by means of 1,10-conjugate addition, giving type I structures.[5] In 2022, Buchwald and Liu disclosed the enantioselective synthesis of type II products via CuH-catalyzed dearomative addition of 3-vinylthiophenes to ketones.[6] Despite these state-of-the-art advances, new synthetic methods based on different synthetic logics are still in demand.

We envisage a skeletal editing strategy[7] for enantioselective and regiodivergent construction of alkylidene-2,3-dihydrothiophenes, by diatomic swap from N−N atoms of 1,2,3-thiadiazoles to two C−C atoms of allenes (Scheme 1C).[8] Although Gevorgyan's,[9] Lee's,[10] and our[11] groups established the prototypes of this skeletal editing with alkynes,[9] alkenes,[10a,d] nitriles,[10b,c] aryl thiocyana-tes[11]—the reactions therein were termed as denitrogenative (3 + 2) transannulations—the asymmetric catalytic versions are very scarce, and the only two reports come from our group.[12] Challenges exist in our synthetic proposal: (1) The highly unsaturated products are susceptible to further aromatization to give nonchiral thiophenes, as demonstrated by our previous work.[13] (2) It is difficult to simultaneously maintain the regiocontrol of the type II and type III products, enantiocontrol of new stereocenters, and the *Z*- or *E*-geometric control of exocyclic C=C bonds of both products in excellent levels. Herein, by adoption of a ligand-controlled tactic (Scheme 1D), the above challenges are well addressed. Our work provides a novel synthetic access to enantiopure 3-dimensional alkylidene-2,3-dihydrothiophenes, which might be used as a potential pharmacore in future drug development campaigns.

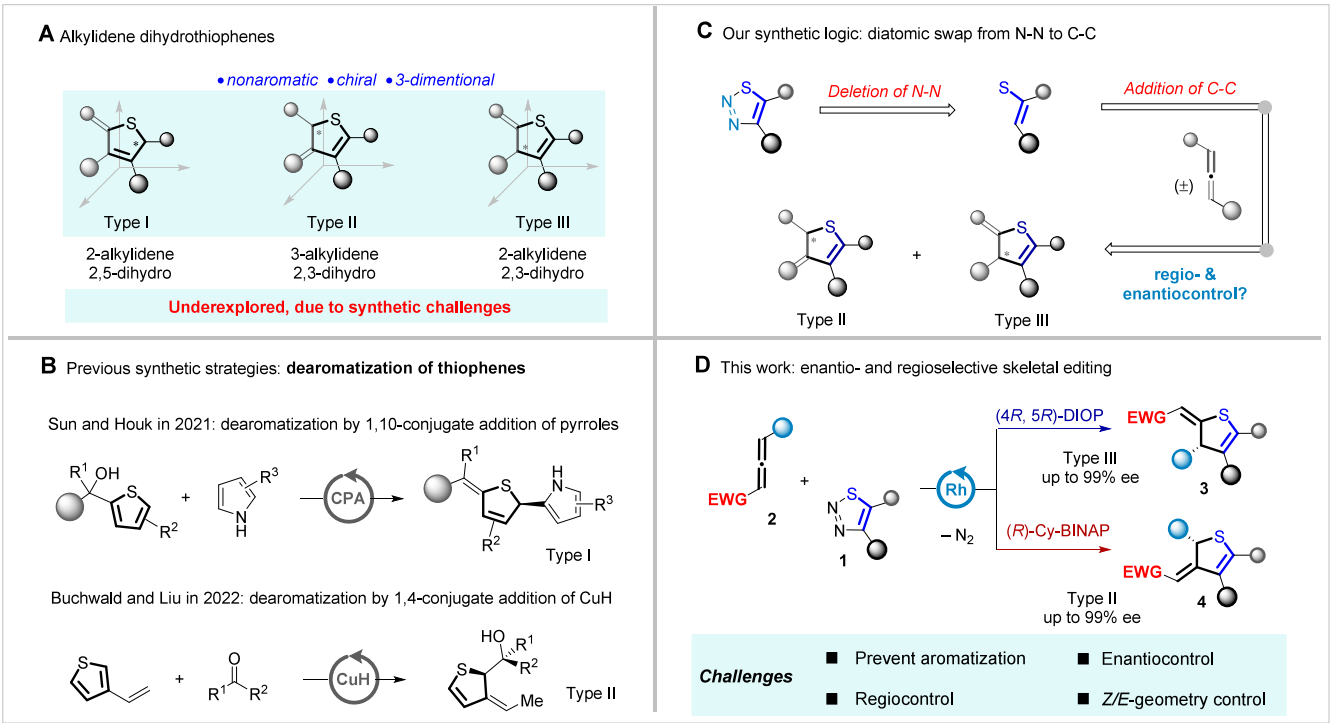
We first optimized the chiral ligands and other reaction parameters, revealing two sets of optimal conditions (see Tables S1 and S2 in Supporting Information): with (4*R*, 5*R*)-DIOP enantiopure 2-alkylidene-2,3-dihydrothiophenes are exclusively furnished (Conditions A), while with (*R*)-Cy-BINAP the regio-reversed 3-alkylidene-2,3-dihydrothiophenes were enantioselectively synthesized (Conditions B). We suggest that the steric hindrance of the ligands controls the regioselectivity. A plausible mechanism is proposed (see Scheme S1 in Supporting

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Scheme 1. Background and Our Strategy for Synthesis of Alkylidene Dihydrothiophenes



Information),[14−16] and theoretical investigations on the origins of regio- and enantiocontrol are underway.

We first evaluated the synthesis of enantiopure type III products **3** (Figure 1A). The reactions showed a broad substrate scope and functional group tolerance. The skeletal editing of 5-aryl-substituted 1,2,3-thiadiazole-4-carboxylates (**1a**−**1o**) proceeded smoothly, regardless of the functionalities on the aryl rings, furnishing **3aA**−**3oA** in 57−95% yields and 96−99% ee. The other regioisomers were formed in such a small amount that they could not be isolated. Well tolerated functional groups include halogens (**3fA**−**3iA**, **3jA**), cyano (**3lA**), nitro (**3kA**), and fur-2-yl (**3mA**), and thiophen-2-yl (**3nA**). Aromatization occurred for **3eA**, accounting for the low yield (32%). However, shortening the reaction time to 80 min significantly improved the yield to 92% (96% ee). In addition, 5-methyl, 5-chloro, and 5-bromo thiadiazoles were also demonstrated as viable substrates, converting to desired products in 90−94% ee (**3pA**−**3rA**). The absolute configuration of **3rA** was unambiguously assigned by single-crystal X-ray diffraction (CCDC: 2404905). The ester group on the 4-position can also be diversified. For example, the reaction of 3-bromopropyl 1,2,3-thiadiazole-4-carboxylate delivered **3sA** in 93% yield and 86% ee, with the labile C(sp^3)−Br bond remaining untouched. Replacement of the ester group with secondary amide slightly lowered the yields, whereas improved the enantioselectivity to 99% ee, as demonstrated by the enantioselective formation of **3tA**−**3zA**.

A number of allenes were also assessed under Conditions A. Diversifying the ester group ((±)-**2H**−**2O**) or replacing the methyl of (±)-**2A** with substituted phenyls ((±)-**2B**−**2F**) and propyl ((±)-**2G**) did not affect the regio- or enantioselectivities. Desired products were obtained in 44−99% yields and 90−97% ee. In addition, a sulfonyl allene was also amenable, giving **3bO** in 90% yield with 90% ee.

By changing (4*R*,5*R*)-DIOP to (*R*)-Cy-BINAP (Conditions B), we accomplished the regio-reversed synthesis of type II products (Figure 1B). Desired products **4aA**−**4dA**, **4gA**, **4hA**, and **4jA**−**4lA** were produced in 50−76% yields and 92−99% ee, with regioselective ratios varying from 3:1 to 17:1. Apparently, the regioselectivity is quite sensitive to the electronic properties of the aromatic substituents. The reactions of four additional allenes (**4bH**, **4aB**, **4aD**, **4aE**) gave excellent enantiocontrol (>99% ee) and almost unchanged regioselectivity (4.5:1 or 4:1 rr). The other regioisomers can be separated by column chromatography. For instance, **3aA** was isolated in 8% yield (96% ee). Notably, the substrate scope under Conditions B is a bit limited (see Scheme S2 in the Supporting Information). For example, 5-methyl or 5-bromo-1,2,3-thiadiazoles failed to undergo the skeletal editing and remained intact.

The scalability was demonstrated by the 1 mmol scale reaction, in which **3aA** was obtained in 89% yield and 96% ee (Scheme 2A). Oxidation of **3aA** with excess *m*-CPBA or reduction with $NaBH_4$−$CoCl_2$ led to sulfone **5** or *trans*-dihydrothiophene **6** in good yields without erosion of the enantiopurity (Scheme 2B). It deserves mention that the exocyclic C═C bond is more susceptible to reduction than the endocyclic one. Deuterium-labeled allene **2E**-*d* afforded deuterium-incorporated product **3aE**-*d* in 90% yield and 95% ee (Scheme 2C). The deuterium of the starting material was almost all retained in the product.

Our protocol can be applied to the late-stage diversification of natural and pharmaceutically relevant molecules (Scheme 2D). Five 1,2,3-thiadiazoles derived from D-galactopyranose (**7a**), cholesterol (**7b**), stigmasterol (**7c**), diosgenin (**7d**), and pregnenolone (**7e**) were transformed efficiently, giving **8aA**−**8eA** in 56−85% yields and 95.5:4.5−98:2 dr. These highly diastereo- and enantiopure structures may arguably serve as potential lead compounds in drug discovery. Dehydroabietylamine and oleylamine were also late-stage modified, giving **8fA**

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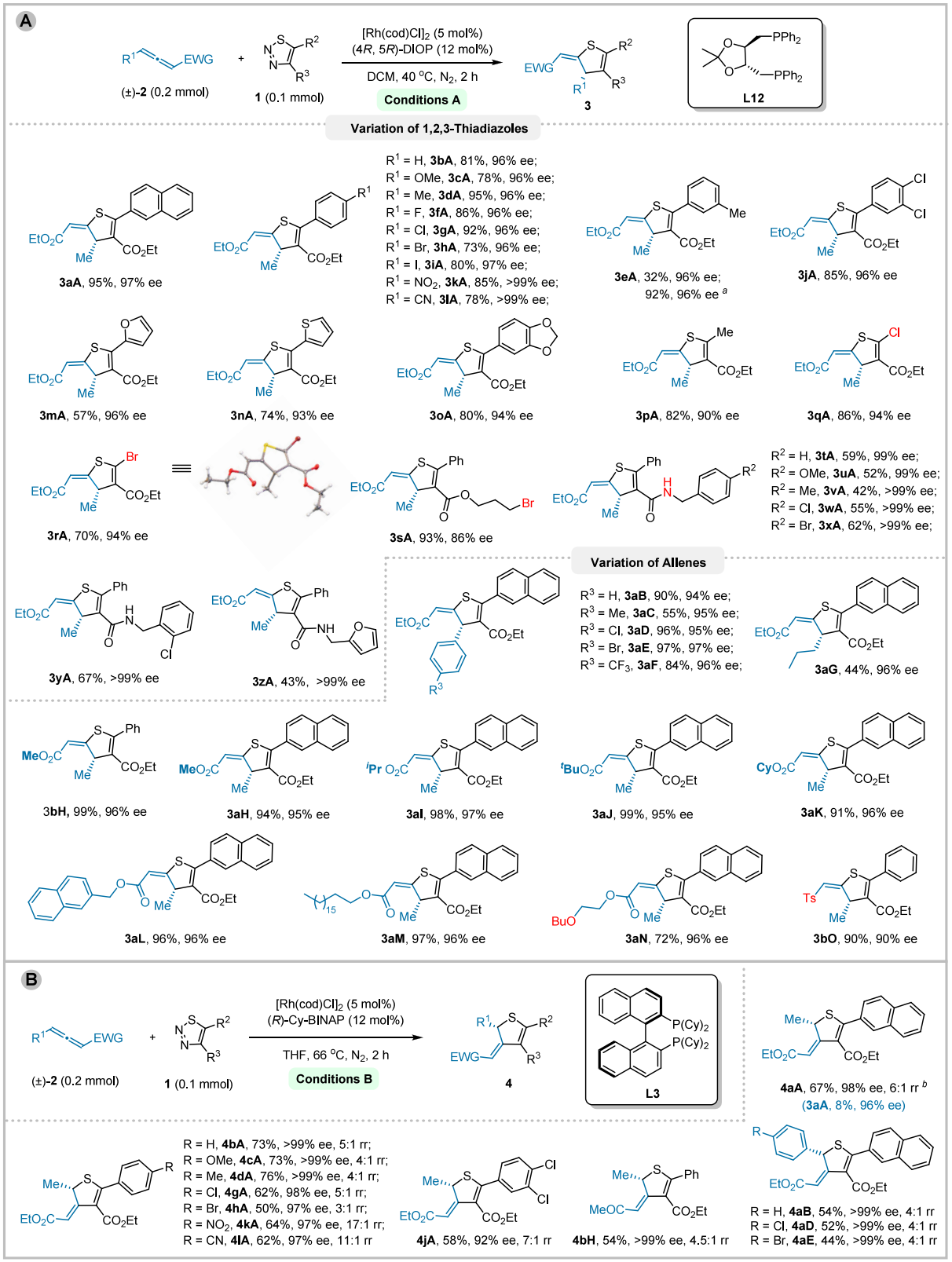


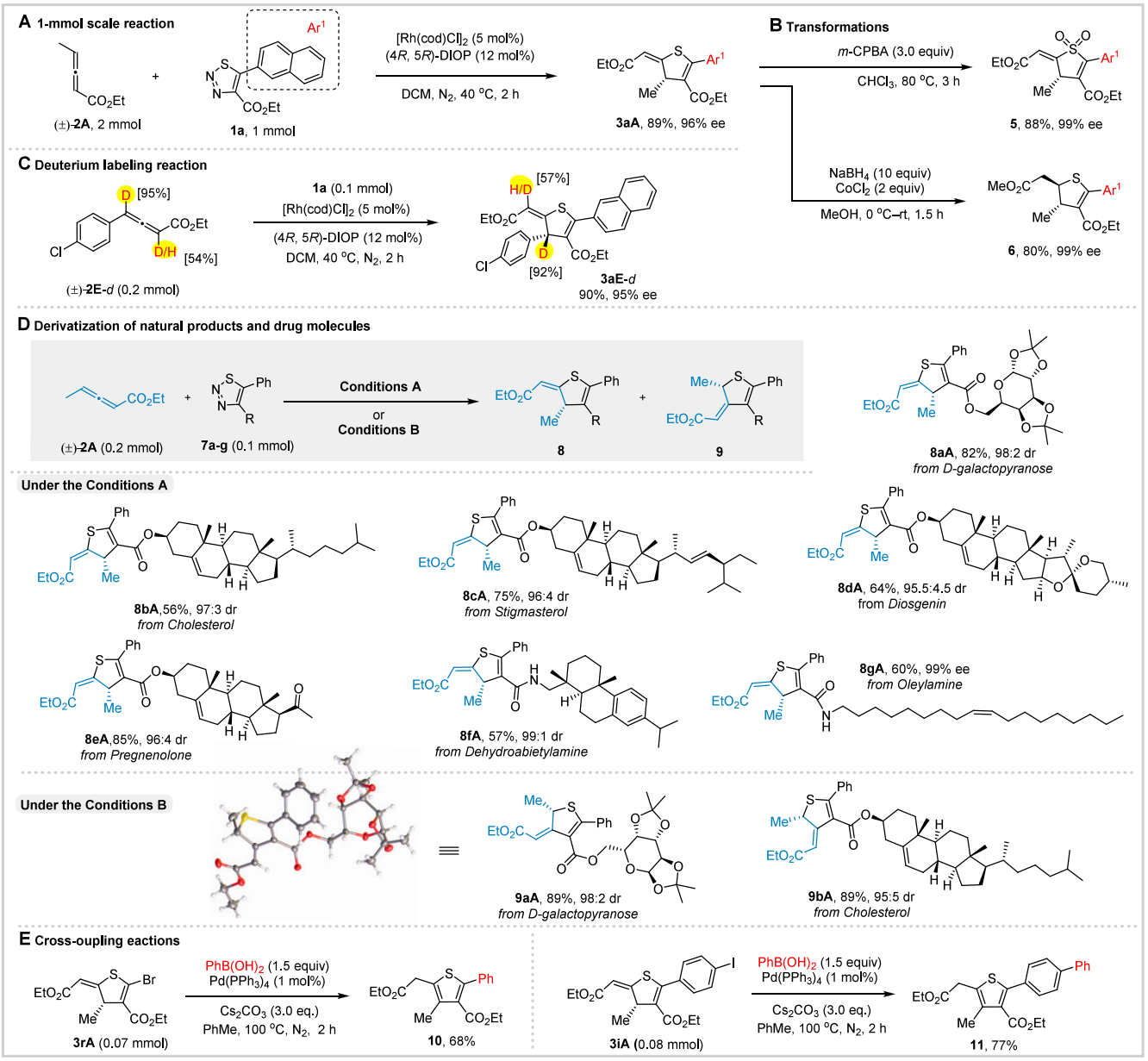
Figure 1. Substrate Scope. [a]The reaction time was shortened to 80 min. [b]All of the rr values are determined by ^{1}H NMR of the crude reaction mixtures.

in 57% yield, 99:1 dr and **8gA** in 60% yield, 99% ee, respectively. On the other hand, the regio-reversed skeletal transformation can also be applied in the structural modification of natural products, as exemplified by the synthesis of **9aA** and **9bA** from D-galactopyranose and cholesterol. These reactions in the presence of complex structures further expand the functionality tolerance.

The bromine and iodine atoms of **3rA** and **3iA** allow for Suzuki coupling reactions, producing thiophenes **10** and **11** in

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Org. Lett. 2025, 27, 4337−4342

Scheme 2. Synthetic Applications and Control Experiments[a]



[a]The dr values are determined by HPLC.

68% and 77% yields, respectively (Scheme 2E). Aromatization took place, probably due to basic conditions.

In conclusion, we have developed a strategically novel rhodium-catalyzed and ligand-controlled regiodivergent and enantioselective synthetic route to alkylidene-2,3-dihydrothiophenes through diatomic skeletal editing of 1,2,3-thiadiazoles with allenoates. Both types of products are synthesized in up to 99% yield and up to 99% ee. Late-stage structural modifications of complex natural and drug molecules are performed, rendering chiral and 3-dimensional chemical spaces with higher levels of molecular complexity. It is anticipated that our synthetic routes to these underexplored enantiopure 3-dimensional alkylidene-2,3-dihydrothiophenes will offer compelling opportunities and provide novel lead compounds for future drug discoveries and developments.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article, in its Supporting Information, and openly available in Dryad at 10.5061/dryad.sxksn03dz.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c01067.

Results on reaction condition optimizations, plausible mechanism, experimental procedures, X-raycrystallographic data for **3rA** and **9aA**, additional Figures S1–S4, Schemes S1–S12, Tables S1, S2, ^{1}H and ^{13}C NMR and HPLC spectroscopic data and spectra, additional references (PDF)

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Accession Codes

Deposition Numbers 2404905 and 2411572 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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